μ is the cause of ν, but we can infer that $\mu \approx \nu$. Why?

Francisco Gutiérrez

1 Introduction and Context

In statistical learning theory, a fundamental challenge is understanding how well a finite sample reflects the underlying population or data-generating distribution. This relationship is often conceptualized using the bin model:

- μ represents the unknown target probability (e.g., the true out-of-sample error E_{out} or the proportion of red marbles in a bin).
- ν represents the observed sample frequency (e.g., the in-sample training error E_{in} or the fraction of red marbles drawn in N trials).

To formalize how close the sample frequency ν is to the true probability μ, we utilize **Hoeffding's Inequality**:

$$\Pr[|\nu - \mu| \geq \epsilon] \leq 2e^{-2N\epsilon^2} \tag{1}$$

Where:

- $\epsilon > 0$ is the tolerance margin for the error between ν and μ.
- N is the sample size.
- $2e^{-2N\epsilon^2}$ provides an upper bound on the probability that ν deviates from μ by more than ϵ.

2 Answering the Question

Question: *μ is the cause of ν, but we can infer that $\mu \approx \nu$. Why?*

1. Causal Direction vs. Inferential Direction

Physically and probabilistically, μ is indeed the generative cause of ν. The true parameter μ governs the distribution from which sample draws are generated, producing an observed sample fraction ν. In forward probability, one calculates the likelihood of observing a specific outcome ν given a known parameter μ.

However, in machine learning and statistics, we face the inverse problem: μ is unknown, and only ν can be directly measured from data.

2. Symmetry of the Absolute Difference

The mathematical validity of inferring $\mu \approx \nu$ stems directly from the algebraic symmetry of the absolute difference in Hoeffding's Inequality:

$$|\nu - \mu| = |\mu - \nu| \tag{2}$$

Because the inequality bounds the magnitude of the difference regardless of direction, the statement:

"The probability that the sample mean ν deviates from μ by at least ϵ is small"

is mathematically identical to:

"The probability that the true parameter μ deviates from the sample mean ν by at least ϵ is small."

3. Exponential Decay and High-Probability Bounds

Because N appears in the negative exponent of $2e^{-2N\epsilon^2}$, as N grows large, the probability of a large deviation drops exponentially toward zero.

Consequently, with high probability $(1 - 2e^{-2N\epsilon^2})$, the unknown target parameter μ is trapped within the interval $[\nu - \epsilon, \nu + \epsilon]$. Therefore, observing ν allows us to legitimately infer that $\mu \approx \nu$ for a sufficiently large sample size N.